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Dyslipidemia in children and adolescents: Definition, screening, and diagnosis

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INTRODUCTION

Dyslipidemias are disorders of lipoprotein metabolism that result in the following abnormalities:

- High total cholesterol (TC)
- High low-density lipoprotein cholesterol (LDL-C)
- High non-high-density lipoprotein cholesterol (non-HDL-C)
- High triglycerides
- Low HDL-C

In adults, dyslipidemia is an established risk factor for atherosclerotic cardiovascular disease (ASCVD), and correcting dyslipidemia reduces the risk of ASCVD. Dyslipidemia often begins in childhood and adolescence. Identifying children with dyslipidemia and successfully improving their lipid profile may reduce their risk of accelerated atherosclerosis and premature ASCVD.

The definition of pediatric dyslipidemia and the approach to screening, evaluation, and diagnosis of lipid disorders in children will be reviewed here. Management of pediatric dyslipidemia is discussed separately. (See "[Dyslipidemia in children and adolescents: Management](#)".)

Other related topics include:

- (See "[Familial hypercholesterolemia in children](#)".)
- (See "[Overview of risk factors for development of atherosclerosis and early cardiovascular disease in childhood](#)".)
- (See "[Overview of the management of the child or adolescent at risk for atherosclerosis](#)".)
- (See "[Inherited disorders of LDL-cholesterol metabolism other than familial hypercholesterolemia](#)".)
- (See "[Screening for lipid disorders in adults](#)".)

DEFINITION

Normative values — Normal lipid and lipoprotein values in children vary by age and sex ([table 1](#)) [1,2]. Normative values are derived from population-based data from the Lipid Research Clinical Prevalence Study, which obtained fasting lipoprotein profiles from 15,626 children (age range 0 to 19 years) between 1972 and 1976 [3], and from the United States National Health and Nutrition Examination Surveys (NHANES), which collected lipid levels in 7000 children from 1988 to 1994 [4-7].

Lipid levels change with normal growth and maturation. Lipoproteins are very low in cord blood at birth and rise slowly in the first two years of life. Higher cholesterol levels are seen in breastfed babies related to the higher saturated fat content of breast milk [8]. Lipid levels are relatively stable from two years of age until adolescence. **During puberty, total cholesterol (TC) and low-density lipoprotein cholesterol (LDL-C) levels decrease with increasing age, before rising in the late teenage years. Males experience a decrease in high-density lipoprotein (HDL) levels during late puberty, whereas HDL levels remain stable in females until menopause.**

Definition of pediatric dyslipidemia — Based on the above normative data, cutoff points are used to delineate lipid values as "acceptable," "borderline," and "abnormal," as shown in the table ([table 2](#)).

These definitions are consistent with guidelines of the National Heart, Lung, and Blood Institute, the American Academy of Pediatrics, and the American Heart Association/American College of Cardiology [1,9-11]. However, it should be noted that these cutoff points have not been validated as accurate predictors for accelerated atherosclerosis or atherosclerotic cardiovascular disease (ASCVD) events.

PREVALENCE

In the United States, approximately 20 percent of children (age 6 to 19 years) have adverse levels of one or more lipid value [2,12-14]. The prevalence of adverse lipid levels increases with age, with 15 percent of children aged 6 to 11 years and 25 percent of adolescents aged 12 to 19 years having at least one adverse level [14].

The prevalence of specific abnormalities are as follows (note that a child may have more than one abnormality) [14]:

- Elevated total cholesterol (TC; ≥ 200 mg/dL [5.2 mmol/L]) – 7.1 percent
- Elevated low-density lipoprotein cholesterol (LDL-C; ≥ 130 mg/dL [3.4 mmol/L]) – 6.4 percent
- Elevated non-high-density lipoprotein cholesterol (non-HDL-C) levels (≥ 145 mg/dL [3.8 mmol/L]) – 6.4 percent
- Elevated triglyceride (≥ 130 mg/dL [1.5 mmol/L]) – 10.2 percent
- Low HDL-C (< 40 mg/dL [1.0 mmol/L]) – 12.1 percent

Lipid levels are impacted by obesity. In an analysis of the United States National Health and Nutrition Examination Survey (NHANES) data from 1996 to 2006, the likelihood of adverse lipid values was higher in adolescents with greater body mass index (BMI). The prevalence among youths who were healthy weight, overweight (BMI 85th to 95th percentile), and obese (BMI $> 95^{\text{th}}$ percentile) were 14, 22, and 43 percent, respectively [15]. However, lipid abnormalities are also common in youth with normal weight. In an analysis of 12- to 19-year-

olds enrolled in NHANES from 1988 to 2016, approximately 52 percent of those with high TC, 38 percent of those with low HDL-C, and 41 percent of youth with high non-HDL-C had a BMI in the normal weight range [16].

Reported rates of pediatric dyslipidemia in other countries are generally comparable to those in the US, including similar relationships to obesity [17,18].

Lipid levels have improved over time. In an analysis of NHANES data from 1999 and 2016, favorable temporal trends were observed in mean lipid levels and in the distribution of ideal and adverse lipid levels among youths aged 6 to 19 years [14]. Mean TC level declined from 164 mg/dL in 1999-2000 to 155 mg/dL in 2015-2016; mean HDL-C level increased from 52.5 mg/dL in 2007-2008 to 55.0 mg/dL in 2015-2016. Trends were generally consistent across racial/ethnic groups and BMI categories. Despite these favorable trends, it was estimated at the end of the study period that only one-half of youths in the United States had all lipids at ideal levels and approximately one in five had at least one adverse level.

ETIOLOGY

The etiology of dyslipidemia can be categorized as follows [19]. In some patients, dyslipidemia may be caused by more than one of these mechanisms.

- Dietary causes – Excessive dietary intake of saturated and trans fats is an important cause or contributor to elevated low-density lipoprotein and excessive intake of refined carbohydrates and simple sugars raises triglyceride levels. (See "Dyslipidemia in children and adolescents: Management", section on 'Dietary modification'.)
- Secondary causes – Many specific diseases and conditions are associated with dyslipidemia. Common secondary causes include obesity, type 2 diabetes mellitus, and nephrotic syndrome. Other secondary causes of dyslipidemia are summarized in the table and are discussed in greater detail separately (table 3). (See "Secondary causes of dyslipidemia".)
- Genetic causes – This includes monogenic and polygenic defects (table 4):

- Monogenic conditions include **familial hypercholesterolemia (FH)**, **familial defective apolipoprotein B or PCSK9**, and **familial hypertriglyceridemia**. (See ["Familial hypercholesterolemia in children"](#) and ["Inherited disorders of LDL-cholesterol metabolism other than familial hypercholesterolemia"](#).)
- Some patients may have a clinical phenotype similar to that of FH but without a single mutation of sufficient pathogenicity to produce it. Such patients likely have multiple gene variants, each of which makes a small independent contribution. These patients are said to have polygenic hypercholesterolemia. (See ["Inherited disorders of LDL-cholesterol metabolism other than familial hypercholesterolemia"](#), section on 'Polygenic hypercholesterolemia'.)

RATIONALE FOR LIPID SCREENING

Benefits of screening — Screening for lipid disorders in childhood is based on the rationale that early identification and control of pediatric dyslipidemia will reduce the risk and severity of atherosclerotic cardiovascular disease (ASCVD) in adulthood [1]. Lipid disorders are clinically silent in the vast majority of cases, and selective screening alone (ie, screening only children with a positive family history) fails to identify a substantial number of children with lipid disorders.

- **Early identification of individuals with FH** – The rationale for universal screening is based, in large part, on the possibility of identifying and treating the greatest number of individuals with familial hypercholesterolemia (FH), a group at high risk for significant morbidity and early mortality [20]. (See ["Familial hypercholesterolemia in children"](#).)
- **Prevention of ASCVD** – There are no randomized controlled trials evaluating the long-term effectiveness of screening for and treating dyslipidemia in childhood, and few data are available on the cost-effectiveness of pediatric lipid screening [21]. The evidence supporting the potential benefits of lipid screening in children comes from studies demonstrating links between pediatric dyslipidemia and atherosclerosis. These data are described below. Studies supporting the efficacy of early treatment of pediatric dyslipidemia are discussed separately. (See ["Dyslipidemia in children and adolescents"](#).)

[Management](#)", section on 'Dosing and efficacy' and ["Familial hypercholesterolemia in children"](#), section on 'Management'.)

Dyslipidemia often begins in childhood and adolescence. Pediatric dyslipidemia contributes to early atherosclerosis and, by extrapolation, to premature ASCVD [1,10]. Moreover, in the high-risk subset of children with severe dyslipidemia due to FH, treatment reduces the risk of cardiovascular events. (See ["Familial hypercholesterolemia in children"](#).)

Evidence for the development of atherosclerosis in childhood includes autopsy studies showing atherosclerotic changes in the young and noninvasive, indirect data in children and adolescents showing vascular changes that commonly precede adult ASCVD. This evidence is discussed in greater detail separately. (See ["Overview of risk factors for development of atherosclerosis and early cardiovascular disease in childhood"](#), section on 'Atherosclerotic changes in childhood'.)

Pediatric lipid disorders often track into adulthood. Approximately one-half of children with abnormal serum lipoprotein values continue to have elevated lipid levels in adulthood [1,19,22,23]. Children with severely elevated lipid values consistent with FH show stronger tracking to adulthood. Cumulative exposure to dyslipidemia appears to be associated with cardiovascular risk later in life. (See ["Familial hypercholesterolemia in children"](#) and ["Familial hypercholesterolemia in adults: Overview"](#).)

Children's lipid values correlate with those of adult family members, and children with dyslipidemia are more commonly found in families with ASCVD [24].

Lifetime low low-density lipoprotein cholesterol (LDL-C) resulting from genetic defects (eg, *PCSK9* gene polymorphisms associated with loss of function) is associated with low lifetime risk of coronary artery disease, while lifetime high LDL-C due to gain-of-function mutations is associated with high ASCVD risk. (See ["Familial hypercholesterolemia in adults: Overview"](#), section on 'Mutations in the *PCSK9* gene'.)

- **Family history: Insensitive predictor** – A positive family history of premature ASCVD (ie, heart attack, treated angina, interventions for coronary artery disease, stroke, or sudden cardiac death in a male parent or sibling before 55 years of age or a female

parent or sibling before 65 years of age) doubles the risk of ASCVD in children and is a well-established ASCVD risk factor in adults [25,26]. However, when used as the sole basis for selective screening in children, positive family history is an insensitive predictor of dyslipidemia.

Selective screening based on family history alone misses a considerable number (30 to 60 percent) of children with dyslipidemia [1,27,28]. In a population-based study of >20,000 5th grade children, the prevalence of abnormal LDL-C levels was similar in children with or without a positive family history [29].

The widespread use of statin therapy in adults for primary prevention of ASCVD has lowered the rate of clinical cardiovascular events, which may explain why family history of premature ASCVD is less predictive of pediatric dyslipidemia than was previously thought.

Despite its limitations as the sole basis for screening, family history is an important consideration, and children with a positive family history should undergo selective screening for dyslipidemia earlier than the recommended universal screening timeframe. A detailed family history is particularly important when evaluating children with possible FH. (See '[Approach to screening](#)' below and "[Familial hypercholesterolemia in children](#)".)

Harms of screening — In the absence of direct long-term data demonstrating that lipid screening in childhood leads to reduced ASCVD in adulthood, some experts have raised concerns that screening may identify a large number of patients who may be harmed by further diagnostic testing and/or initiation of drug therapy with uncertain benefit [30-34].

It is estimated that 100,000 children <17 years and >400,000 adolescents and young adults aged 17 to 21 years in the United States would qualify for statin therapy based on the National Heart, Lung, and Blood Institute screening recommendations [35-37]. Statin treatment may be beneficial for these patients. However, the potential risks of statin therapy in this population must also be considered. **Rare but important side effects of statin therapy in adults include rhabdomyolysis and new-onset type 2 diabetes mellitus. Adverse effects of statin therapy in children are rare and are discussed separately.** (See "[Dyslipidemia in](#)

children and adolescents: Management", section on 'Adverse effects' and "Statins: Actions, side effects, and administration", section on 'Side effects'.)

More data are needed to address these important questions and to determine whether there is overall benefit for universal screening of pediatric dyslipidemia.

APPROACH TO SCREENING

The following sections outline our suggested approach to screening for dyslipidemia during childhood and adolescence. We use an approach that combines universal, selective, and cascade screening strategies.

Our approach is generally consistent with the [guidelines](#) of the National Heart, Lung, and Blood Institute, the American Academy of Pediatrics, the American Heart Association, and the American College of Cardiology [1,9-11]. However, this approach is not universally accepted. A 2023 report of the United States Preventive Services Task Force, after review of the available literature, concluded that data were insufficient to recommend for or against routine screening in children and adolescents [27]. The 2008 National Institute for Health and Care Excellence guidelines and the 2015 consensus statement of the European Atherosclerosis Society recommend selective screening and cascade screening to identify children and adults with FH [38,39]. These and other published guidelines can be accessed through the links provided below. (See '[Society guideline links](#)' below.)

Who should be screened — We suggest using a combined approach of universal and selective screening based on the age of the child and the presence of underlying risk factors that increase the risk of early atherosclerotic cardiovascular disease (ASCVD) ([algorithm 1](#)).

- **Children with ASCVD risk factors** – For children with one or more risk factors for premature ASCVD ([table 5](#)), we suggest regular screening for dyslipidemia. Screening typically begins at the age when the ASCVD risk factor is first identified. In the case of a family history of hypercholesterolemia or premature atherosclerotic ASCVD, screening typically begins after the age of two years. The interval of subsequent surveillance testing is tailored to the individual's risk profile (typically every one to three years) and continues so long as the risk factor(s) persist.

- **Children with close relatives with known monogenic lipid disorders** – For children with a first-degree (parent, sibling) or second-degree (grandparent, aunt, uncle) relative with a known genetic lipid disorder, such as familial hypercholesterolemia, we recommend cascade testing either with lipid testing as below, or if a molecular diagnosis has been made, testing for the specific identified mutation. This issue is discussed in greater detail separately. (See ["Familial hypercholesterolemia in children", section on 'Testing family members'](#).)
- **Children without ASCVD risk factors** – For children without any risk factors for premature ASCVD, we suggest routine screening for dyslipidemia twice during childhood and late adolescence. The first screening should be performed between age 9 and 11 years and the second between age 17 and 21 years. Between ages 12 and 16 years, screening is **not** recommended for children without ASCVD risk factors, because changes in lipid levels that normally occur during puberty decrease the sensitivity and specificity for predicting adult low-density lipoprotein cholesterol (LDL-C) levels and increase false-negative results in this age group.

Choice of screening test — Lipid levels can be measured either in a fasting or nonfasting state. Treatment decisions are generally made based on fasting lipid profiles; however, screening can be accomplished with nonfasting lipid levels. If the initial screen is performed with a nonfasting test and the result is abnormal, a follow-up fasting lipid profile should be obtained. At least two fasting profiles should be used to guide clinical decision-making, including pharmacologic therapy. (See ['Follow-up after screening'](#) below.)

- **Lipid profile** – The full lipid profile includes:
 - Total cholesterol (TC), measured directly
 - High-density lipoprotein cholesterol (HDL-C), measured directly
 - Triglycerides (TG), measured directly – Since TG levels are influenced by recent food intake, treatment decisions should be based on fasting levels
 - LDL-C, calculated using the Friedewald formula ([calculator 1](#)) or the Martin method [40,41]

Nonfasting lipid levels often are more practical as a screening test for pediatric patients

because many patients and families find it challenging to comply with fasting. Differences in TC and HDL-C measurements between the fasting or nonfasting state are small and clinically insignificant [42]. The nonfasting non-HDL-C level (which is calculated based on the difference between TC and HDL-C) appears to be a sensitive screening test for dyslipidemia in children [43]. Non-HDL-C includes all cholesterol present in lipoprotein particles that are considered atherogenic, including LDL-C, lipoprotein(a), intermediate-density lipoprotein, and very-low-density lipoprotein.

In an analysis from the Bogalusa study, non-HDL-C was at least as good a predictor as other lipid tests (ie, LDL-C, TC, HDL-C, and the ratio of TC:HDL-C) for predicting increased carotid intima-media thickness (an indirect marker for atherosclerosis) [44]. Another report from the Bogalusa study found that abnormal levels of childhood non-HDL-C persisted into adulthood and were predictive of adult dyslipidemia independent of baseline body mass index (BMI) and BMI changes over 27 years [43].

In the Pathobiological Determinants of Atherosclerosis in Youth study (PDAY study), non-HDL-C and HDL-C levels were the best lipid predictors of pathologic atherosclerotic lesions in autopsies of children who had died from noncardiac causes [45].

Of note, the Friedewald formula, which is used to calculate LDL-C levels in the fasting lipid profile is valid only if the TG level is <400 mg/dL (<4.5 mmol/L). The Sampson equation can be used for patients with TG levels up to 800 mg/dL [46]. In patients with more pronounced hypertriglyceridemia, LDL-C levels can be measured directly (direct LDL). We do not routinely use direct LDL as a screening test because nonHDL performs well, is readily available, and is less costly [47,48]. (See "Measurement of blood lipids and lipoproteins", section on 'Measurement'.)

FOLLOW-UP AFTER SCREENING

After the initial screening test, repeat testing is required for those with abnormal results to confirm the diagnosis of pediatric dyslipidemia and determine the need for intervention ([algorithm 2](#)). Very high cholesterol values (LDL-C level of ≥ 250 mg/dL [6.5 mmol/L]) are suggestive of familial hypercholesterolemia (FH) and generally warrant additional evaluation

and therapy. (See ["Familial hypercholesterolemia in children"](#), section on 'Evaluation'.)

Normal screen — Patients with "acceptable" values on lipid screening ([table 2](#)) do not require any further evaluation. They should continue to undergo regular cardiovascular health assessments, including lipid screening at the intervals outlined above ([algorithm 1](#)). (See ["Approach to screening"](#) above.)

Borderline screen — For patients with borderline lipid results, recommendations for a heart-healthy lifestyle should be reinforced as for all children and adolescents. Further follow-up and testing is tailored to the specific clinical scenario. Important considerations include the age of the patient and underlying medical conditions or other risk factors ([table 5](#)). In most cases, it is reasonable to repeat testing in one year. (See ["Dyslipidemia in children and adolescents: Management"](#), section on 'Dietary modification' and ["Pediatric prevention of adult cardiovascular disease: Promoting a healthy lifestyle and identifying at-risk children"](#).)

Abnormal screen — Patients with adversely high or low values on the initial lipid screening test ([table 2](#)) should have confirmatory testing performed with a fasting lipid profile. If dyslipidemia is confirmed, patients should undergo evaluation for secondary causes of dyslipidemia. (See ["Confirmatory testing"](#) below and ["Secondary causes of hypercholesterolemia"](#) below.)

Confirmatory testing — [The diagnosis of dyslipidemia requires confirmation testing with fasting lipid profiles obtained on two separate occasions two weeks to three months apart \[1\].](#) Decisions regarding the need for pharmacotherapy should be based on the results of at least two low-density lipoprotein cholesterol (LDL-C) values derived from fasting lipid profiles. Counseling regarding dietary and other lifestyle changes should begin with the first abnormal test. (See ["Dyslipidemia in children and adolescents: Management"](#).)

Repeated testing is required because there is considerable intraindividual variability. In a report from the Bogalusa study, children with LDL-C levels between 160 and 189 mg/dL (4.1 and 4.9 mmol/L) demonstrated an average decrease in LDL-C by 21 mg/dL (0.5 mmol/L) at the next examination and, among those with levels ≥ 190 mg/dL (≥ 4.9 mmol/L), the average decrease was 34 mg/dL (0.9 mmol/L) [49]. Patients and families should be counseled in lifestyle modification to promote normalization of values between testing. (See ["Dyslipidemia](#)

in children and adolescents: Management", section on 'Heart-healthy lifestyle'.)

Secondary causes of hypercholesterolemia — Patients with confirmed dyslipidemia should be evaluated for secondary causes of hypercholesterolemia, which include ([table 3](#)):

- Diabetes mellitus, particularly uncontrolled disease (see "[Complications and screening in children and adolescents with type 1 diabetes mellitus](#)", section on 'Cardiovascular disease')
- Nephrotic syndrome (see "[Clinical manifestations, diagnosis, and evaluation of nephrotic syndrome in children](#)")
- Hypothyroidism (see "[Acquired hypothyroidism in childhood and adolescence](#)")
- Pregnancy (see "[Approach to evaluating pregnant patients with elevated liver biochemical and function tests](#)")
- Hepatic disease (see "[Nonalcoholic fatty liver disease in children and adolescents](#)")
- Drugs (eg, alcohol, glucocorticoids, [isotretinoin](#), antiretrovirals) (see "[Major side effects of systemic glucocorticoids](#)", section on 'Cardiovascular effects' and "[Oral isotretinoin therapy for acne vulgaris](#)", section on 'Hyperlipidemia')

Many of these conditions can be identified through the history and physical examination. Additional laboratory evaluation may include serum alanine aminotransferase, serum albumin, blood glucose level, renal function tests (ie, blood urea nitrogen and creatinine), serum thyroid-stimulating hormone, and urine human chorionic gonadotropin screen, if clinically indicated.

Very high cholesterol (LDL-C 250 or higher) — [Children with LDL-C \$\geq\$ 250 mg/dL \(6.5 mmol/L\) should be referred directly to a pediatric lipid specialist for further evaluation and management.](#) Referral may also be helpful for other patients if LDL levels remain elevated ($\geq$ 160 mg/dL despite adequate lifestyle modification counseling). Markedly elevated cholesterol levels are associated with a high likelihood of FH and almost certainly require pharmacotherapy. (See "[Familial hypercholesterolemia in children](#)".)

DIFFERENCES IN PEDIATRIC VERSUS ADULT APPROACHES

There are two key differences between the pediatric approach to lipid screening, described above (see ['Approach to screening'](#) above), and adult primary prevention approaches, which are discussed separately (see ["Screening for lipid disorders in adults"](#)):

- **Global versus individual risk assessment** – Adult primary prevention treatment recommendations are based in part on calculating a patient's individual baseline risk for atherosclerotic cardiovascular disease (ASCVD) by using validated risk scores (eg, the Framingham and the Reynolds risk scores, and the American College of Cardiology/American Heart Association guidelines [50]), which were developed from robust outcome data. (See ["Cardiovascular disease risk assessment for primary prevention: Risk calculators"](#) and ["Low-density lipoprotein cholesterol-lowering therapy in the primary prevention of cardiovascular disease"](#), section on 'CVD Risk Assessment'.)

Scoring systems for estimating ASCVD risk in children are not available, because similar robust outcomes data are generally lacking. Therefore, it is not possible to accurately estimate a child's individual risk for ASCVD.

- **Lifetime versus 10-year risk** – There are two approaches to assessing ASCVD risk in adults: the 10-year risk and the "lifetime" (eg, 30-year) risk [51]. The lifetime risk approach is more relevant to pediatric practice since the 10-year risk for children and adolescents is exceedingly low. Adult risk assessment scores, which were developed based on the ASCVD risk in a general population, should not be used in individuals with heterozygous familial hypercholesterolemia (FH) who have a lifetime risk for ASCVD events that is not adequately captured by these scores [52,53].

The differences in these two approaches may result in a discrepancy of recommendations regarding statin therapy as older adolescents transition their care from pediatric to adult primary care clinicians [37]. The "right" approach to treating lipid disorders in young adults remains to be determined.

SOCIETY GUIDELINE LINKS

Links to society and government-sponsored guidelines from selected countries and regions around the world are provided separately. (See ["Society guideline links: Lipid disorders and atherosclerosis in children"](#).)

SUMMARY AND RECOMMENDATIONS

- **Definitions** – Dyslipidemias are disorders of lipoprotein metabolism that result in one or more of the following abnormalities ([table 2](#)) (see ['Introduction'](#) above and ['Definition of pediatric dyslipidemia'](#) above):
 - Total cholesterol (TC) >200 mg/dL (5.2 mmol/L)
 - Low-density lipoprotein cholesterol (LDL-C) >130 mg/dL (3.4 mmol/L)
 - Triglycerides >100 mg/dL (1.1 mmol/L) in children <10 years and >130 mg/dL (1.5 mmol/L) in children >10 years
 - High-density lipoprotein cholesterol (HDL-C) <40 mg/dL (1.0 mmol/L)
 - Non-HDL-C >145 mg/dL (3.8 mmol/L)
- **Etiology** – Pediatric dyslipidemia can be caused by any of the following mechanisms, alone or in combination (see ['Etiology'](#) above):
 - Excessive dietary intake of saturated and trans fats (see ["Dyslipidemia in children and adolescents: Management"](#), section on ['Dietary modification'](#))
 - Secondary causes such as obesity, type 2 diabetes mellitus, and nephrotic syndrome ([table 3](#)) (see ["Secondary causes of dyslipidemia"](#))
 - Genetic causes, including monogenetic and polygenic defects (see ["Familial hypercholesterolemia in children"](#) and ["Inherited disorders of LDL-cholesterol metabolism other than familial hypercholesterolemia"](#))
- **Rationale for screening** – Dyslipidemia often begins in childhood and adolescence and tracks into adulthood. Pediatric dyslipidemia contributes to early atherosclerosis and, by extrapolation, to premature atherosclerotic cardiovascular disease (ASCVD). In the

high-risk subset of children with severe dyslipidemia due to familial hypercholesterolemia (FH), treatment reduces the risk of cardiovascular events. Screening for lipid disorders in childhood is based on the rationale that early identification and control of pediatric dyslipidemia will reduce the risk and severity of ASCVD in adulthood. (See '[Benefits of screening](#)' above.)

- **Approach to screening** – Our approach to screening for dyslipidemia in children and adolescents uses a strategy of combined age-based universal and selective screening ([algorithm 1](#)) (see '[Approach to screening](#)' above):
 - **Children with risk factors** – For children with one or more risk factors for premature ASCVD ([table 5](#)), we suggest regular screening for dyslipidemia (**Grade 2C**). Screening typically begins at the age when the ASCVD risk factor is first identified (generally, not earlier than age two years, unless there is clinical suspicion for homozygous FH is suspected). The interval of testing is tailored to the individual's risk profile (typically, every one to three years) and continues so long as the risk factor(s) persist. (See '[Who should be screened](#)' above.)
 - **Children without risk factors** – For children without any risk factors for premature ASCVD, we suggest routine screening for dyslipidemia twice during childhood and late adolescence (**Grade 2C**). The first screening should be performed between age 9 and 11 years and the second between age 17 and 21 years. Screening should **not** be performed at age 12 to 16 years in children without ASCVD risk factors, because changes in lipid levels that normally occur during puberty decrease the sensitivity and specificity of screening. (See '[Who should be screened](#)' above.)
 - **Choice of screening test** – Lipid screening can be performed with a full fasting lipid profile or with nonfasting lipid levels (with the latter, non-HDL-C is calculated based on the TC and HDL-C levels). Abnormal results require confirmatory testing. (See '[Choice of screening test](#)' above.)
- **Follow-up after screening**
 - **Repeat lipid profile** – For patients with abnormal results, repeat testing is performed with a fasting lipid profile to confirm the diagnosis and determine the

need for intervention ([algorithm 2](#)). (See '[Confirmatory testing](#)' above.)

- **Evaluation for secondary causes** – Patients with confirmed dyslipidemia should be evaluated for secondary causes of hypercholesterolemia, which include diabetes mellitus, nephrotic syndrome, hypothyroidism, pregnancy, hepatic disease, and certain medications ([table 3](#)). (See '[Secondary causes of hypercholesterolemia](#)' above.)
- **Referral** – Children with LDL-C ≥ 250 mg/dL (6.5 mmol/L) should be referred to a pediatric lipid specialist since markedly elevated cholesterol levels are associated with a high likelihood of FH and almost certainly require pharmacotherapy. Referral may also be necessary for children with less severe lipid profiles if LDL-C levels remain elevated (≥ 160 mg/dL) despite adequate lifestyle modification counseling. (See '[Very high cholesterol \(LDL-C 250 or higher\)](#)' above and "[Familial hypercholesterolemia in children](#)".)

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GRAPHICS

Lipid and lipoprotein distributions in subjects aged 5 to 19 years

	Males			Females		
	5 to 9 years	10 to 14 years	15 to 19 years	5 to 9 years	10 to 14 years	15 to 19 years
Total cholesterol, mg/dL						
50 th percentile	153	161	152	164	159	157
75 th percentile	168	173	168	177	171	176
90 th percentile	183	191	183	189	191	198
95 th percentile	186	201	191	197	205	208
Triglyceride, mg/dL						
50 th percentile	48	58	68	57	68	64
75 th percentile	58	74	88	74	85	85
90 th percentile	70	94	125	103	104	112
95 th percentile	85	111	143	120	120	126
LDL, mg/dL						
50 th percentile	90	94	93	98	94	93
75 th percentile	103	109	109	115	110	110
90 th percentile	117	123	123	125	126	129

95 th percentile	129	133	130	140	136	137
HDL, mg/dL						
5 th percentile	38	37	30	36	37	35
10 th percentile	43	40	34	38	40	38
25 th percentile	49	46	39	48	45	43
50 th percentile	55	55	46	52	52	51

Normal fasting lipoprotein profiles from the United States National Health and Nutrition Examination Surveys (NHANES).

LDL: low-density lipoprotein; HDL: high-density lipoprotein.

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Acceptable, borderline, and abnormal plasma lipid, lipoprotein, and apolipoprotein concentrations for children and adolescents

Category	Acceptable mg/dL (mmol/L)	Borderline mg/dL (mmol/L)	High mg/dL (mmol/L)
TC	<170 (4.4)	170 to 199 (4.4 to 5.2)	≥200 (5.2)
LDL-C	<110 (2.8)	110 to 129 (2.8 to 3.3)	≥130 (3.4)
Non-HDL-C	<120 (3.1)	120 to 144 (3.1 to 3.7)	≥145 (3.8)
ApoB	<90 (2.3)	90 to 109 (2.3 to 2.8)	≥110 (2.8)
TG			
• 0 to 9 years	<75 (0.8)	75 to 99 (0.8 to 1.1)	≥100 (1.1)
• 10 to 19 years	<90 (1 mmol/L)	90 to 129 (1 to 1.5)	≥130 (1.5)
Category	Acceptable mg/dL (mmol/L)	Borderline mg/dL (mmol/L)	Low mg/dL (mmol/L)
HDL-C	>45 (1.2)	40 to 45 (1 to 1.2)	<40 (1)
ApoA-1	>120 (3.1)	115 to 120 (3 to 3.1)	<115 (3)

Values for plasma lipid and lipoprotein levels are from the National Cholesterol Education Program (NCEP) Expert Panel on Cholesterol Levels in Children. Non-HDL-C values from the Bogalusa Heart Study are equivalent to the NCEP Pediatric Panel cut points for LDL-C. Values for plasma ApoB and ApoA-1 are from the National Health and Nutrition Examination Survey III (NHANES III). The threshold points for high and borderline-high values represent approximately the 95th and 75th percentiles, respectively. Low threshold points for HDL-C and ApoA-1 represent approximately the 10th percentile.

TC: total cholesterol; LDL-C: low-density lipoprotein cholesterol; HDL-C: high-density lipoprotein cholesterol; ApoB: apolipoprotein B; TG: triglycerides; ApoA-1: apolipoprotein A-1.

Reproduced from: Daniels SR, Benuck I, Christakis DA, et al. Expert panel on integrated guidelines for cardiovascular health and risk reduction in children and adolescents: Full report, 2011. National Heart Lung and Blood Institute. Available at: http://www.nhlbi.nih.gov/guidelines/cvd_ped/peds_guidelines_full.pdf.

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Causes of secondary dyslipidemia in children

Exogenous
Alcohol
Drug therapy
- Corticosteroids - Isotretinoin - Some oral contraceptives - Select chemotherapeutic agents - Select antiretroviral agents
Endocrine/Metabolic
- Hypothyroidism/hypopituitarism - Diabetes mellitus types 1 and 2 - Pregnancy - Polycystic ovary syndrome - Lipodystrophy - Acute intermittent porphyria
Renal
- Chronic renal disease - Hemolytic uremic syndrome - Nephrotic syndrome
Infectious
- Acute viral/bacterial infection* - HIV infection - Hepatitis
Hepatic
Obstructive liver disease/cholestatic conditions

Biliary cirrhosis
Alagille syndrome
Inflammatory disease
Systemic lupus erythematosus
Juvenile rheumatoid arthritis
Storage disease
Glycogen storage disease
Gaucher disease
Cystine storage disease
Juvenile Tay-Sachs disease
Niemann-Pick disease
Other
Kawasaki disease
Anorexia nervosa
Solid organ transplantation
Childhood cancer survivor
Progeria
Idiopathic hypercalcemia
Klinefelter syndrome
Werner syndrome

* Delay measurement until ≥ 3 weeks postinfection.

Adapted from: Daniels SR, Benuck I, Christakis DA, et al. Expert panel on integrated guidelines for cardiovascular health and risk reduction in children and adolescents: Full report, 2011. National Heart Lung and Blood Institute. Available at: http://www.nhlbi.nih.gov/guidelines/cvd_ped/peds_guidelines_full.pdf.

Graphic 71294 Version 5.0

Characteristic findings of different genetic causes of hypercholesterolemia

Disorder	Estimated prevalence	Characteristic lipid levels*	Clinical features	
			In adults	In children
Heterozygous familial hypercholesterolemia	1/200 to 1/300	▪ Elevated TC and LDL-C ▪ LDL-C 190 to 450 mg/dL[¶] ▪ TG usually normal	▪ Tendon or tuberosus xanthomata ▪ Xanthelasmas ▪ Corneal arcus ▪ Premature CHD	▪ Subclinical atherosclerosis
Homozygous familial hypercholesterolemia	1/1,000,000	▪ Markedly elevated TC and LDL-C ▪ Untreated LDL-C >500 mg/dL ▪ TG usually normal	▪ Extensive xanthomata ▪ Severe and progressive atherosclerotic cardiovascular disease	▪ Cutaneous tendon xanthomata often in the first year ▪ Premature CHD occurs in childhood untreated
FCHL	1 to 2%	▪ Phenotypically heterogeneous ▪ Elevated TC and/or TG ▪ LDL-C/APOB ratio <1.2	▪ Xanthelasma ▪ Corneal arcus ▪ Premature CHD	▪ Clinical manifestations are typically not seen during childhood
Familial hyperapobetalipoproteinemia	<1% Likely a variant of FCHL	▪ APOB >135 mg/dL ▪ LDL-C <160 mg/dL ▪ LDL-C/APOB ratio <1.2	▪ Xanthelasma ▪ Premature CHD ▪ Obesity	▪ Clinical manifestations other than obesity are typically seen during childhood

Polygenic hypercholesterolemia	>25%	▪ Elevated TC and LDL-C ▪ TG usually normal ▪ LDL-C 130-250 mg/dL	▪ Premature CHD ▪ Tendon xanthomata are not seen	▪ Clinical manifest are typical not seen during childhood
Familial dysbetalipoproteinemia	1/5000	▪ Elevated TC and TG ▪ VLDL/TG ratio >0.3	▪ Tuberoeruptive xanthomata ▪ Xanthomata of the palmar creases ▪ Premature CHD	▪ Clinical manifest are typical not seen during childhood

Summary of the clinical features of different lipid disorders.

TC: total cholesterol; LDL-C: low-density lipoprotein cholesterol; TG: triglycerides; CHD: coronary heart disease; LDLR: LDL receptor; APOB: apolipoprotein B; PCSK9: proprotein convertase subtilisin kexin 9; FCHL: familial combined hyperlipidemia; VLDL: very low-density lipoprotein; APOE: apolipoprotein E.

* Lipid levels presented in this table represent the typical ranges seen in each disorder, primarily in affected adults. In most of these disorders, there is considerable variability in these values and lipid levels alone generally cannot be used to confirm or exclude the diagnosis.

¶ In pediatric patients, an LDL-C cutoff value of ≥ 190 mg/dL is generally used to establish a diagnosis of heterozygous FH. However, in patients with a concerning family history (ie, first-degree relatives with hypercholesterolemia and/or early cardiovascular disease), it may be appropriate to use a lower cutoff value (eg, LDL of ≥ 160 mg/dL).

Graphic 66747 Version 5.0

Our approach to lipid screening in children and adolescents

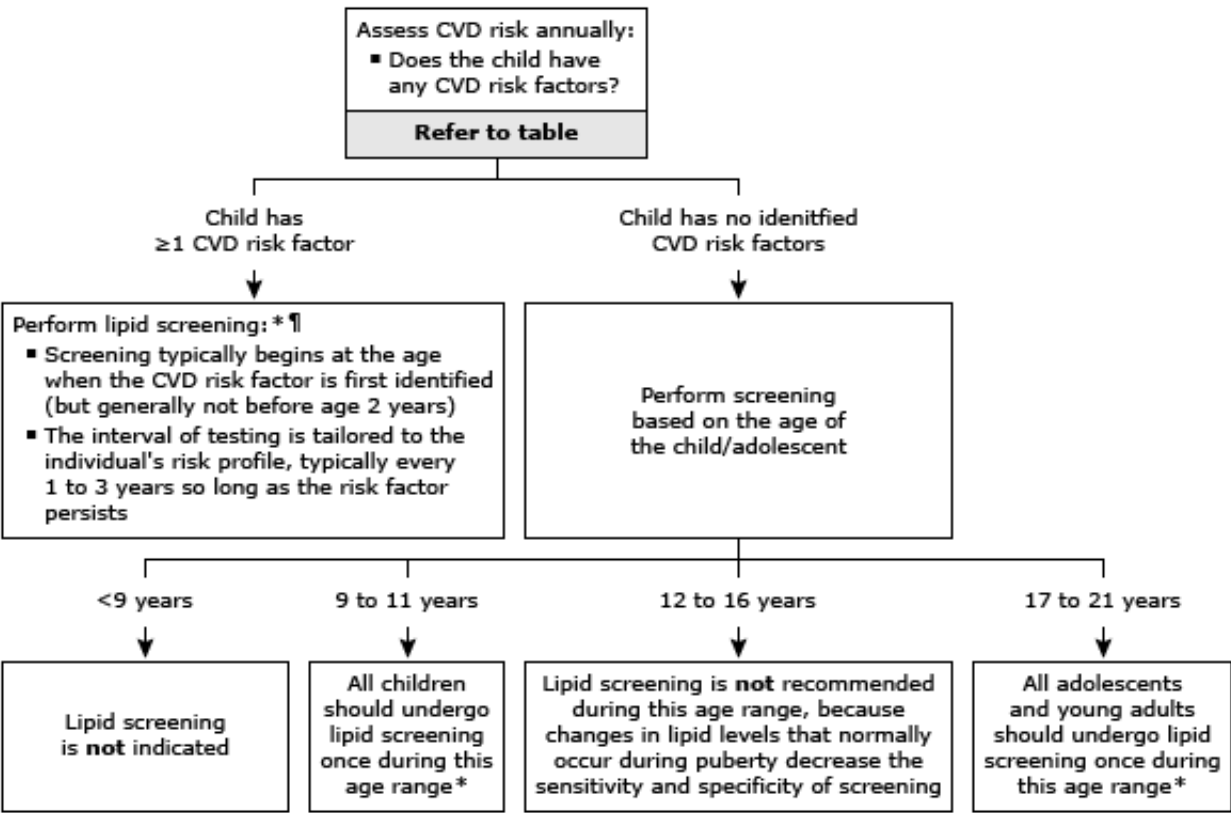


Table: Risk factors for atherosclerosis
Tradition:
▪ Dyslipidemia
▪ Obesity
▪ Diabetes
▪ Hypertension
▪ Family history of premature CVD
▪ Smoke
Other conditions:
▪ Familial hypercholesterolemia
▪ Chronic kidney disease
▪ Kawasaki disease
▪ Childhood obesity
▪ Transplantation
▪ Certain congenital heart diseases, e.g., transposition of the large vessels
▪ Cardiac surgery
▪ Chronic inflammatory diseases, e.g., HIV infection, rheumatoid arthritis
▪ Adolescent idiopathic scoliosis

CVD: cardiovascular disease; CoA: coarctation of the aorta; AS: aortic stenosis; TGA: transposition of the large vessels; HCM: hypertrophic cardiomyopathy; SLE: systemic lupus erythematosus; JIA: juvenile idiopathic arthritis; HDL-C: high-density lipoprotein cholesterol; LDL-C: low-density lipoprotein cholesterol.

* Initial screening can be performed with a fasting or nonfasting lipid profile (with the latter, non-HDL-C is calculated from measured TC and HDL-C levels). Confirmatory testing with a fasting lipid profile should be performed if a screen is positive (ie, TC ≥200 mg/dL, HDL-C <40 mg/dL, LDL-C ≥130 mg/dL, or non-HDL-C ≥145 mg/dL). For additional information on screening and diagnosis of dyslipidemia in children.

¶ These are general suggestions for the timing and interval of screening; they should be tailored to the individual child. For example, for a child with a single traditional risk factor (eg, nonsevere obesity or smoke exposure without other conditions), it is reasonable to begin screening later and repeat it less frequently. By contrast, for a child with other conditions (eg, Kawasaki disease with known coronary artery aneurysm), it is reasonable to perform lipid screening more frequently.

Δ Family history of premature CVD is generally defined as heart attack, treated angina, interventions for coronary artery disease, cardiac death, or ischemic stroke in a male parent or sibling before 55 years of age or a female parent or sibling before 65 years of age.

Adapted from:
1. Expert Panel on Integrated Guidelines for Cardiovascular Health and Risk Reduction in Children and Adolescents, National Heart Lung and Blood Institute (NHLBI). *Guidelines for lipid screening in children and adolescents: Summary report.* Pediatrics. 2011;128(5):e136-151.

2. *de Ferranti SD, Steinberger J, Ameduri R, et al. Cardiovascular risk reduction in high-risk pediatric patients: A scientific state of the science review. Circulation 2019; 139:e603.*
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Graphic 105367 Version 9.0

Risk factors for development of atherosclerosis and early cardiovascular disease in childhood

Traditional risk factors
Dyslipidemia
Obesity
Diabetes mellitus (type 1 or 2)
Hypertension
Family history of premature ASCVD*
Smoke exposure
Other conditions with increased ASCVD risk
Familial hypercholesterolemia
Chronic kidney disease
Kawasaki disease
Childhood cancer
Transplant vasculopathy
Certain congenital heart disease defects (eg, CoA, AS, TGA, congenital coronary artery anomalies)
Cardiomyopathy (eg, HCM)
Chronic inflammatory disorders (eg, SLE, systemic JIA)
HIV infection
Adolescent depressive and bipolar disorders

ASCVD: atherosclerotic cardiovascular disease; CoA: coarctation of the aorta; AS: aortic stenosis; TGA: transposition of the great arteries; HCM: hypertrophic cardiomyopathy; SLE: systemic lupus erythematosus; JIA: juvenile idiopathic arthritis.

* Family history of premature ASCVD is generally defined as one or more of the following before age 55 years in a male parent or sibling or before age 65 years in a female parent or sibling:

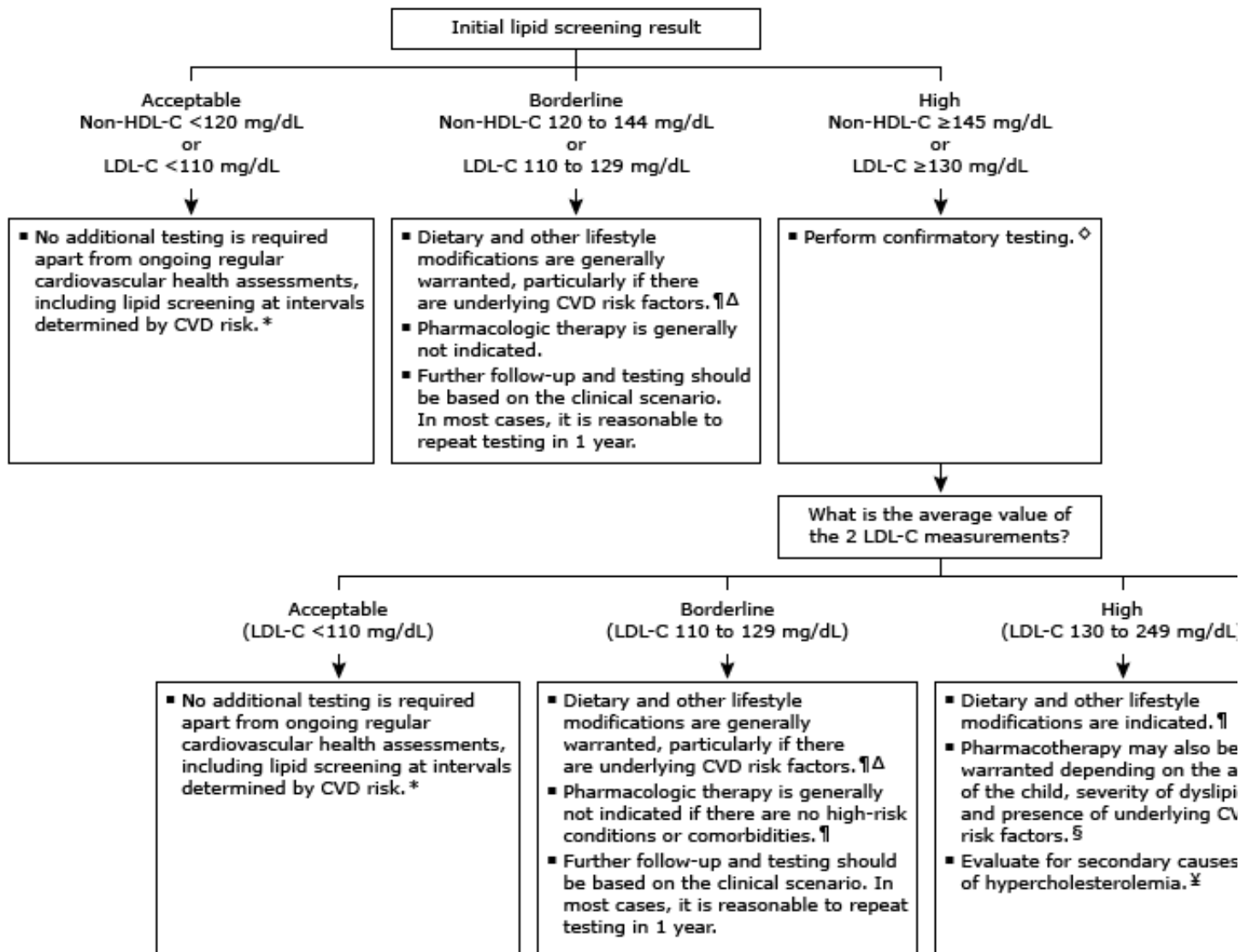
- Heart attack, treated angina, or interventions for coronary artery disease
- Sudden cardiac death
- Ischemic stroke

References:

1. De Ferranti SD, Steinberger J, Ameduri R, et al. Cardiovascular risk reduction in high-risk pediatric patients: A scientific statement from the American Heart Association. *Circulation* 2019; 139:e603.
 2. Expert Panel on Integrated Guidelines for Cardiovascular Health and Risk Reduction in Children and Adolescents, National Heart, Lung, and Blood Institute. Expert panel on integrated guidelines for cardiovascular health and risk reduction in children and adolescents: Summary report. *Pediatrics* 2011; 128 Suppl 5:S213.
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Graphic 109927 Version 7.0

Overview of follow-up of children undergoing lipid screening



LDL-C and non-HDL-C units are mg/dL; divide by 38.67 to convert to mmol/L.

HDL-C: high-density lipoprotein cholesterol; LDL-C: low-density lipoprotein cholesterol; CVD: cardiovascular disease; hypercholesterolemia; SLE: systemic lupus erythematosus; JIA: juvenile idiopathic arthritis; CoA: coarctation of the aorta; HCM: hypertrophic cardiomyopathy.

* For children with underlying risk factors for premature CVD (listed below)Δ, lipid screening begins at the age of 2 years (generally not earlier than age 2 years) with subsequent testing every 1 to 3 years depending on the nature of the risk factors. Without underlying risk factors for premature CVD, routine lipid screening is performed twice during childhood (ages 9 to 11 years and 17 to 21 years) and once after puberty (age 17 to 21 years). Refer to UpToDate topic on dyslipidemia screening in children and adolescents.

¶ Lifestyle modifications may include low saturated fat diet; increased intake of dietary fiber through fruits and vegetables; supplementation with plant stanols and sterols; increasing physical activity; and, in obese children, weight loss.

management of dyslipidemia in children for further details of nonpharmacologic management of hypercholesterolemia. Δ Risk factors for premature CVD in children are similar to those in adults and include hypertension, diabetes, and family history of premature atherosclerotic CVD. In addition, other medical conditions that can be associated with increased risk of early CVD include: chronic kidney disease, Kawasaki disease (with current or past coronary artery aneurysms), childhood cancer, transplant vasculopathy, chronic inflammatory diseases (eg, SLE and systemic JIA), HIV infection, aortic disease (CoA, AS, TGA, congenital coronary artery anomalies), cardiomyopathy (eg, HCM), pulmonary hypertension, and bipolar disorder.

◇ Confirmatory testing consists of 2 fasting lipid profiles obtained 2 weeks to 3 months apart (ie, if the profile was performed as the initial screen, a second fasting lipid profile is obtained 2 weeks to 3 months later; if the profile was using nonfasting non-HDL-C, 2 separate fasting lipid profiles should be performed). Average the 2 results.

§ For further details on the management of hypercholesterolemia and FH, refer to separate UpToDate topics on these subjects.

¥ Secondary causes of hypercholesterolemia include diabetes mellitus, nephrotic syndrome, hypothyroidism, and certain medications (eg, glucocorticoids, antiretrovirals). Refer to the UpToDate topic on screening and management of secondary causes of hypercholesterolemia for further details.

Graphic 121848 Version 1.0

